Computer Network Project

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-Tables:

1.Subnet Table:

Subnet Description	VLAN ID	VLAN Name	Required Number of Hosts	Network Address/CIDR	First Usable Host Address	Last Usable Host Address	Broadcast Address
Main	Vlan 100	Vlan100	61	172.16.0.0/26	172.16.0.1	172.16.0.62	172.16.0.63
R	Vlan 700	Vlan700	31	172.16.0.64/26	172.16.0.65	172.16.0.126	172.16.0.127
Main	Vlan 200	Vlan200	30	172.16.0.128/27	172.16.0.129	172.16.0.158	172.16.0.159
N	Vlan 600	Vlan600	29	172.16.0.160/27	172.16.0.161	172.16.0.190	172.16.0.191
R	Vlan 800	Vlan800	25	172.16.0.192/27	172.16.0.193	172.16.0.222	172.16.0.223
S	Vlan 400	Vlan400	20	172.16.0.224/27	172.16.0.225	172.16.0.254	172.16.0.255
N	Vlan 500	Vlan500	15	172.16.1.0/27	172.16.1.1	172.16.1.30	172.16.1.31
S	Vlan 25	Vlan25	12	172.16.1.32/28	172.16.1.33	172.16.1.46	172.16.1.47
From SW-S to MIU-MIU-GW	-	-	7	172.16.1.48/28	172.16.1.49	172.16.1.62	172.16.1.63
From Main-MLS to MIU-MIU-GW	-	-	2	172.16.1.64/30	172.16.1.65	172.16.1.66	172.16.1.67
From N-MLS to MIU-MIU-GW	-	-	2	172.16.1.68/30	172.16.1.69	172.16.1.70	172.16.1.71
From S-MLS to MIU-MIU-GW	-	-	2	172.16.1.72/30	172.16.1.73	172.16.1.74	172.16.1.75
From R-MLS to MIU-MIU-GW	-	-	2	172.16.1.76/30	172.16.1.77	172.16.1.78	172.16.1.79

2.Addressing Table:

Device	Interface	Address	Subnet Mask	Default Gateway	VLAN ID
Main-MLS	G1/1/1	172.16.1.65	255.255.255.252	-	-
	VLAN 100	172.16.0.1	255.255.255.192	-	Vlan 100
	VLAN 200	172.16.0.129	255.255.255.224	-	Vlan 200
S-MLS	G1/1/1	172.16.1.73	255.255.255.252	-	-
	VLAN 25	172.16.1.33	255.255.255.240	-	Vlan 25
	VLAN 400	172.16.0.225	255.255.255.224	-	Vlan 400
N-MLS	G1/1/1	172.16.1.69	255.255.255.252	-	-
	VLAN 500	172.16.1.1	255.255.255.224	-	Vlan 500
	VLAN 600	172.16.0.161	255.255.255.224	-	Vlan 600
R-MLS	G1/0/5	172.16.1.77	255.255.255.252	-	-
	VLAN 700	172.16.0.65	255.255.255.192	-	Vlan 700
	VLAN 800	172.16.0.193	255.255.255.224	-	Vlan 800
MIU-GW	Gig0/0	209.165.200.226	255.255.255.240	-	-
	Gig0/1	172.16.1.49	255.255.255.240	-	-
	Gig0/0/0	172.16.1.66	255.255.255.252	-	-
	Gig0/1/0	172.16.1.70	255.255.255.252	-	-
	Gig0/2/0	172.16.1.74	255.255.255.252	-	-
	Gig0/3/0	172.16.1.78	255.255.255.252	-	-
ISP	Gig 0/0	209.165.200.225	255.255.255.240	-	-
	Gig 0/1	64.100.1.1	255.255.255.224	-	-
	Gig 0/2	64.100.2.1	255.255.255.224	-	-
Branch-GW	Gig 0/0.2	192.168.2.1	255.255.255.0	-	-
	Gig 0/0.3	192.168.3.1	255.255.255.0	-	-
	Gig 0/1	64.100.1.2	255.255.255.224	-	-
Wireless Home Router	G0/1	64.100.2.2	255.255.255.224	-	-
	Router IP	192.168.10.1	255.255.255.128	64.100.2.1	-

3.Hosts Address Table:

PC-number	VLAN ID	IP Address	Subnet Mask	Default Gateway
MIU				
PC-1	VLAN 100	From DHCP Server	From DHCP Server	From DHCP Server
PC-2	VLAN 100	From DHCP Server	From DHCP Server	From DHCP Server
PC-3	VLAN 200	From DHCP Server	From DHCP Server	From DHCP Server
PC-4	VLAN 25	From DHCP Server	From DHCP Server	From DHCP Server
PC-5	VLAN 25	From DHCP Server	From DHCP Server	From DHCP Server
PC-6	VLAN 400	From DHCP Server	From DHCP Server	From DHCP Server
PC-7	VLAN 500	172.16.1.2	255.255.255.224	172.16.1.1
PC-8	VLAN 600	172.16.0.162	255.255.255.224	172.16.0.161
PC-9	VLAN 700	172.16.0.66	255.255.255.192	172.16.0.65
PC-10	VLAN 800	172.16.0.194	255.255.255.224	172.16.0.193
MIU_ Branch 1				
PC-11	VLAN 2	From DHCP Server	From DHCP Server	From DHCP Server
PC-12	VLAN 3	From DHCP Server	From DHCP Server	From DHCP Server
Wireless Home Network				
Laptop		192.168.10.10	255.255.255.128	192.168.10.1
Tablet		192.168.10.20	255.255.255.128	192.168.10.1
Smartphone		192.168.10.30	255.255.255.128	192.168.10.1
Servers				
DHCP Server		172.16.1.50	255.255.255.240	172.16.1.49
Email Server		172.16.1.51	255.255.255.240	172.16.1.49
Web server		172.16.1.52	255.255.255.240	172.16.1.49
DNS server		172.16.1.53	255.255.255.240	172.16.1.49
NTP and Syslog server		172.16.1.54	255.255.255.240	172.16.1.49

-Configurations and Implementations:

-Devices Configuration and Implementations:

Build the Network and Configure Basic Device Settings and Interface Addressing.

-MIU_GW:

```
MIU-GW#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     209.165.200.226 YES manual up                    up
GigabitEthernet0/1     172.16.1.78     YES manual up                    up
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/0   172.16.1.49     YES manual up                    up
GigabitEthernet0/1/0   172.16.1.66     YES manual up                    up
GigabitEthernet0/2/0   172.16.1.74     YES manual up                    up
GigabitEthernet0/3/0   172.16.1.70     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
MIU-GW#
```

-Main_MLS:

```
Main-MLS#SHOW IP INTERFACE BRIEF
Interface              IP-Address      OK? Method Status                Protocol
Port-channel2          unassigned      YES unset  up                    up
Port-channel3          unassigned      YES unset  up                    up
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  down                  down
GigabitEthernet1/0/6   unassigned      YES unset  down                  down
GigabitEthernet1/0/7   unassigned      YES unset  down                  down
GigabitEthernet1/0/8   unassigned      YES unset  down                  down
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   172.16.1.65     YES manual up                    up
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan100                172.16.0.1      YES manual up                    up
Vlan200                172.16.0.129    YES manual up                    up
Main-MLS#
```

-S_MLS:

```
S-MLS#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
Port-channel2          unassigned      YES unset  up                    up
Port-channel3          unassigned      YES unset  up                    up
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  down                  down
GigabitEthernet1/0/6   unassigned      YES unset  down                  down
GigabitEthernet1/0/7   unassigned      YES unset  down                  down
GigabitEthernet1/0/8   unassigned      YES unset  down                  down
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   172.16.1.73     YES manual up                    up
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan25                 172.16.1.33     YES manual up                    up
Vlan400                172.16.0.225    YES manual up                    up
```

-N_MLS:

```
N-MLS#SHOW IP INT BRIEF
Interface              IP-Address      OK? Method Status                Protocol
Port-channel2          unassigned      YES unset  up                    up
Port-channel3          unassigned      YES unset  up                    up
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  down                  down
GigabitEthernet1/0/6   unassigned      YES unset  down                  down
GigabitEthernet1/0/7   unassigned      YES unset  down                  down
GigabitEthernet1/0/8   unassigned      YES unset  down                  down
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   172.16.1.69     YES manual up                    up
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan500                172.16.1.1      YES manual up                    up
Vlan600                172.16.0.161    YES manual up                    up
```

-R_MLS:

```
R-MLS#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
Port-channel2          unassigned      YES unset  up                    up
Port-channel3          unassigned      YES unset  up                    up
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   172.16.1.77     YES manual up                    up
GigabitEthernet1/0/6   unassigned      YES unset  down                  down
GigabitEthernet1/0/7   unassigned      YES unset  down                  down
GigabitEthernet1/0/8   unassigned      YES unset  down                  down
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan700                172.16.0.65     YES manual up                    up
Vlan800                172.16.0.193    YES manual up                    up
```

-MIU_Branch:

```
Branch_GW#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.2   192.168.2.1     YES manual up                    up
GigabitEthernet0/0.3   192.168.3.1     YES manual up                    up
GigabitEthernet0/1     64.100.1.2      YES manual up                    up
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

-ISP:

```
Password:
ISP#SHOW IP INT BRIEF
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     209.165.200.225 YES manual up                    up
GigabitEthernet0/1     64.100.1.1      YES manual up                    up
GigabitEthernet0/2     64.100.2.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
ISP#
```

Layer 2 network configuration and set up basic host support.

-Main Building:

-Basic Set-up:

```
Main-MLS#show run
Building configuration...

Current configuration : 2404 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Main-MLS
!
!
no profinet
enable secret 5 $1$mERr$rieJ5nEP8XfGN.NvipeTy1

no ip cef
ip routing
!
no ipv6 cef
!

no ip domain-lookup
!
!
spanning-tree mode pvst

banner motd ^CAUTHORIZED ACCESS ONLY^C
!
!
!
!
logging trap debugging
logging 172.16.1.54
line con 0
 password 7 080C657B584B5643475D5B
 login
!
line aux 0
!
line vty 0 4
 login
!
!
!
ntp server 172.16.1.54
!
end
```

-VLANS:

```
Main-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
100  VLAN0100                         active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Main-SW1#
```

```
Main-SW2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
100  VLAN0100                         active    Fa0/3
200  VLAN0200                         active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Main-SW2#
```

```
Main-MLS#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/5, Gig1/0/6, Gig1/0/7,
Gig1/0/8

                                                Gig1/0/9, Gig1/0/10, Gig1/0/11,
Gig1/0/12

                                                Gig1/0/13, Gig1/0/14, Gig1/0/15,
Gig1/0/16

                                                Gig1/0/17, Gig1/0/18, Gig1/0/19,
Gig1/0/20

                                                Gig1/0/21, Gig1/0/22, Gig1/0/23,
Gig1/0/24

                                                Gig1/1/2, Gig1/1/3, Gig1/1/4
100  100                              active
200  200                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

-Trunking:

```
Main-SW1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po2         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po2         1-1005

Port        Vlans allowed and active in management domain
Po1         1,100
Po2         1,100

Port        Vlans in spanning tree forwarding state and not pruned
Po1         none
Po2         1,100
```

```
Main-SW2#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po1         1,100,200
Po3         1,100,200

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,100,200
Po3         1,100,200
```

```
Main-MLS#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po2         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po2         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po2         1,100,200
Po3         1,100,200

Port        Vlans in spanning tree forwarding state and not pruned
Po2         1,100,200
Po3         1,100,200
```

-Eatherchannels:

```
Main-MLS#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

2      Po2(SU)           LACP   Gig1/0/1(P) Gig1/0/2(P)
3      Po3(SU)           LACP   Gig1/0/3(P) Gig1/0/4(P)
```

```
Main-SW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P)  Fa0/2(P)
2      Po2(SU)           LACP   Gig0/1(P)  Gig0/2(P)
```

```
Main-SW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P)  Fa0/2(P)
3      Po3(SU)           LACP   Gig0/1(P)  Gig0/2(P)
```

-S Building:

-Basic Set-up:

```
S-MLS#show run
Building configuration...

Current configuration : 2609 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname S-MLS
!
!
no profinet
enable secret 5 $1$mERr$rieJ5nEP8XfGN.NvipeTy1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
.
no ip domain-lookup
!
!
spanning-tree mode pvst
!

ip classless
!
ip flow-export version 9
!
!
!
banner motd ^CAUTHORIZED ACCESS ONLY^C
!
!
!
!
logging trap debugging
logging 172.16.1.54
line con 0
 password 7 080C657B584B5643475D5B
 login
!
line aux 0
!
line vty 0 4
 login
!
!
!
ntp server 172.16.1.54
!
end
```

-VLANS:

```
S-MLS#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/5, Gig1/0/6, Gig1/0/7,
Gig1/0/8
                                                Gig1/0/9, Gig1/0/10, Gig1/0/11,
Gig1/0/12
                                                Gig1/0/13, Gig1/0/14, Gig1/0/15,
Gig1/0/16
                                                Gig1/0/17, Gig1/0/18, Gig1/0/19,
Gig1/0/20
                                                Gig1/0/21, Gig1/0/22, Gig1/0/23,
Gig1/0/24
                                                Gig1/1/2, Gig1/1/3, Gig1/1/4
25   VLAN25                           active
400  VLAN400                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S-MLS#
```

```
S-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
25   VLAN25                           active    Fa0/3
400  VLAN400                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S-SW1#
```

```
S-SW2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
25   VLAN25                           active    Fa0/3
400  VLAN400                          active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S-SW2#
```

-Trunking:

```
S-MLS#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po2         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po2         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po2         1,25,400
Po3         1,25,400

Port        Vlans in spanning tree forwarding state and not pruned
Po2         1,25,400
Po3         1,25,400
```

```
S-SW1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po2         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po2         1-1005

Port        Vlans allowed and active in management domain
Po1         1,25,400
Po2         1,25,400

Port        Vlans in spanning tree forwarding state and not pruned
Po1         none
Po2         1,25,400

S-SW1#
```

```
S-SW2#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po1         1,25,400
Po3         1,25,400

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,25,400
Po3         1,25,400
```

-Eatherchannels:

```
S-MLS#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

2      Po2(SU)           LACP   Gig1/0/1(P) Gig1/0/2(P)
```

```
S-SW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P) Fa0/2(P)
2      Po2(SU)           LACP   Gig0/1(P) Gig0/2(P)
```

```
S-SW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P) Fa0/2(P)
3      Po3(SU)           LACP   Gig0/1(P) Gig0/2(P)
```

-N Building:

-Basic Set-up:

```
N-MLS#show run
Building configuration...

Current configuration : 2314 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname N-MLS
!
!
no profinet
enable secret 5 $1$mERr$rieJ5nEP8XfGN.NvipeTy1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef

:
no ip domain-lookup
!
!
spanning-tree mode pvst
!

ip classless
ip route 0.0.0.0 0.0.0.0 172.16.1.70
!
ip flow-export version 9
!
!
!
banner motd ^CAUTHORIZED ACCESS ONLY^C
!
!
!
!
logging trap debugging
logging 172.16.1.54
line con 0
 password 7 080C657B584B5643475D5B
 login
!
line aux 0
!
line vty 0 4
 login
!
!
!
ntp server 172.16.1.54
!
end
```

-VLANS:

```
N-MLS#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/5, Gig1/0/6, Gig1/0/7,
Gig1/0/8
                                                Gig1/0/9, Gig1/0/10, Gig1/0/11,
Gig1/0/12
                                                Gig1/0/13, Gig1/0/14, Gig1/0/15,
Gig1/0/16
                                                Gig1/0/17, Gig1/0/18, Gig1/0/19,
Gig1/0/20
                                                Gig1/0/21, Gig1/0/22, Gig1/0/23,
Gig1/0/24
                                                Gig1/1/2, Gig1/1/3, Gig1/1/4
500  VLAN500                          active
600  VLAN600                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
N-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
500  VLAN500                          active    Fa0/3
600  VLAN600                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
N-SW1#
```

```
N-SW2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
500  VLAN500                          active
600  VLAN600                          active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
N-SW2#
```

-Trunking:

```
N-MLS#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po2         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po2         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po2         1,500,600
Po3         1,500,600

Port        Vlans in spanning tree forwarding state and not pruned
Po2         1,500,600
Po3         1,500,600
```

```
N-SW1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po2         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po2         1-1005

Port        Vlans allowed and active in management domain
Po1         1,500,600
Po2         1,500,600

Port        Vlans in spanning tree forwarding state and not pruned
Po1         none
Po2         1,500,600

N-SW1#
```

```
N-SW2#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po1         1,500,600
Po3         1,500,600

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,500,600
Po3         1,500,600

N-SW2#
```

-Eatherchannels:

```
N-MLS#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

2      Po2(SU)           LACP   Gig1/0/1(P) Gig1/0/2(P)
3      Po3(SU)           LACP   Gig1/0/3(P) Gig1/0/4(P)
N-MLS#
```

```
N-SW1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P)  Fa0/2(P)
2      Po2(SU)           LACP   Gig0/1(P)  Gig0/2(P)
N-SW1#
```

```
N-SW2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P)  Fa0/2(P)
3      Po3(SU)           LACP   Gig0/1(P)  Gig0/2(P)
N-SW2#
```

-MIU_Branch 1:

-Basic Set-up:

```
Branch_GW#SHOW RUN
Building configuration...

Current configuration : 2284 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
security passwords min-length 10
!
hostname Branch_GW
!
!
!
enable secret 5 $1$mERr$rieJ5nEP8XfGN.NvipeTy1
!
!
ip dhcp excluded-address 192.168.2.1 192.168.2.5
ip dhcp excluded-address 192.168.3.1 192.168.3.5
!
ip dhcp pool B-LAN@
ip dhcp pool B-LAN2
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 172.16.1.53
ip dhcp pool B-LAN3
 network 192.168.3.0 255.255.255.0
 default-router 192.168.3.1
 dns-server 172.16.1.53
!
!
!
ip cef
no ipv6 cef
!
!
!
username admin secret 5 $1$mERr$7QPLAIhw9fatFu382mE0P1
!
!
```

```
!
license udi pid CISCO2911/K9 sn FTX152408IY-
license boot module c2900 technology-package securityk9
!
!
!
crypto isakmp policy 10
 encr aes 256
 authentication pre-share
 group 5
!
crypto isakmp key VPNpa55 address 64.100.1.2
crypto isakmp key VPNpa55 address 209.165.200.226
!
!
!
crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha-hmac
!
crypto map VPN-MAP 10 ipsec-isakmp
 set peer 209.165.200.226
 set transform-set VPN-SET
 match address VPN-ACL
!
!
!
!
!
!
no ip domain-lookup
ip domain-name miu.edu.eg
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
```

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.2
 encapsulation dot1Q 2
 ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/0.3
 encapsulation dot1Q 3
 ip address 192.168.3.1 255.255.255.0
!
interface GigabitEthernet0/1
 ip address 64.100.1.2 255.255.255.224
 duplex auto
 speed auto
 crypto map VPN-MAP
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 64.100.1.1
!
ip flow-export version 9
!
!
ip access-list extended VPN-ACL
 permit ip 192.168.2.0 0.0.0.255 172.16.0.0 0.0.255.255
 permit ip 192.168.3.0 0.0.0.255 172.16.0.0 0.0.255.255
 permit ip 192.168.0.0 0.0.255.255 172.16.0.0 0.0.255.255
!
banner motd ^CAUTHORIZED ACCESS ONLY^C
```


```
banner motd ^CAUTHORIZED ACCESS ONLY^C
```


```
!
!
!
!
logging trap debugging
logging 172.16.1.54
line con 0
 password 7 080C657B584B5643475D5B
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
!
!
ntp server 172.16.1.54
!
end
```

-VLANS:

```
B_S1#SHOW VLAN BRIEF

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
2    VLAN2                            active    Fa0/3
3    VLAN3                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
B_S2#SHOW VLAN BRIEF

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
2    VLAN2                            active
3    VLAN3                            active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

-Trunking:

```
B_S1#SHOW VLAN BRIEF

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
2    VLAN2                            active    Fa0/3
3    VLAN3                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
B_S1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Po1         1,2,3
Gig0/1      1,2,3

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,2,3
Gig0/1      1,2,3
```

```
B_S2#SHOW VLAN BRIEF

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
2    VLAN2                            active    
3    VLAN3                            active    Fa0/3
1002 fddi-default                     active    
1003 token-ring-default               active    
1004 fddinet-default                  active    
1005 trnet-default                    active    
B_S2#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005

Port        Vlans allowed and active in management domain
Po1         1,2,3

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,2,3
```

-Etherchannel:

```
B_S1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P) Fa0/2(P)
B_S1#
```

-show ip route:

```
Branch_GW#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 64.100.1.1 to network 0.0.0.0

     64.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       64.100.1.0/27 is directly connected, GigabitEthernet0/1
L       64.100.1.2/32 is directly connected, GigabitEthernet0/1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0.2
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0.2
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0.3
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0.3
S*   0.0.0.0/0 [1/0] via 64.100.1.1

Branch_GW#
```

-R Building:

-Basic Set-up:

```
R-MLS#show run
Building configuration...

Current configuration : 2384 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R-MLS
!
!
no profinet
enable secret 5 $1$mERr$rieJ5nEP8XfGN.NvipeTy1
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
```

```
no ip domain-lookup
!
!
spanning-tree mode pvst
!
```

```
ip classless
ip route 0.0.0.0 0.0.0.0 172.16.1.78
!
ip flow-export version 9
!
!
!
banner motd ^CAUTHORIZED ACCESS ONLY^C
!
!
!
!
logging trap debugging
logging 172.16.1.54
line con 0
 password 7 080C657B584B5643475D5B
 login
!
line aux 0
!
line vty 0 4
 login
!
!
!
ntp server 172.16.1.54
!
end
```

-VLANS:

```
R-MLS#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/6, Gig1/0/7, Gig1/0/8,
Gig1/0/9
                                                Gig1/0/10, Gig1/0/11, Gig1/0/12,
Gig1/0/13
                                                Gig1/0/14, Gig1/0/15, Gig1/0/16,
Gig1/0/17
                                                Gig1/0/18, Gig1/0/19, Gig1/0/20,
Gig1/0/21
                                                Gig1/0/22, Gig1/0/23, Gig1/0/24,
Gig1/1/1
                                                Gig1/1/2, Gig1/1/3, Gig1/1/4
700  700                              active
800  800                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R-MLS#
```

```
R-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Po3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24
500  VLAN0500                         active
700  VLAN700                          active    Fa0/3
800  VLAN800                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R-SW1#
```

```
R-SW2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
600  VLAN0600                         active
700  VLAN700                          active
800  VLAN800                          active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R-SW2#
```

-Trunking:

```
R-MLS#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po2         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po2         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po2         1,700,800
Po3         1,700,800

Port        Vlans in spanning tree forwarding state and not pruned
Po2         1,700,800
Po3         1,700,800

R-MLS#
```

```
R-SW1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po2         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po2         1-1005

Port        Vlans allowed and active in management domain
Po1         1,500,700,800
Po2         1,500,700,800

Port        Vlans in spanning tree forwarding state and not pruned
Po1         500
Po2         1,500,700,800
```

```
R-SW2#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Po3         on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Po3         1-1005

Port        Vlans allowed and active in management domain
Po1         1,600,700,800
Po3         1,600,700,800

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,600,700,800
Po3         1,600,700,800

R-SW2#
```

-Eatherchannels:

```
R-MLS#show etherchannel summary
Flags:  D - down         P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

2      Po2(SU)           LACP   Gig1/0/1(P) Gig1/0/2(P)
3      Po3(SU)           LACP   Gig1/0/3(P) Gig1/0/4(P)
R-MLS#
```

```
R-SW1#show etherchannel summary
Flags:  D - down         P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 3
Number of aggregators:           3

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P)  Fa0/2(P)
2      Po2(SU)           LACP   Gig0/1(P)  Gig0/2(P)
3      Po3(SD)           -
R-SW1#
```

```
R-SW2#show etherchannel summary
Flags:  D - down         P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/1(P)  Fa0/2(P)
3      Po3(SU)           LACP   Gig0/1(P)  Gig0/2(P)
R-SW2#
```

-Routing Protocols:

Configure IPv4 routing protocols.

-Main_GW:

```
MIU-GW#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.200.225 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 18 subnets, 5 masks
O       172.16.0.0/26 [110/2] via 172.16.1.65, 00:02:30, GigabitEthernet0/1/0
D       172.16.0.64/26 [90/25625856] via 172.16.1.77, 00:03:19, GigabitEthernet0/1
O       172.16.0.128/27 [110/2] via 172.16.1.65, 00:02:30, GigabitEthernet0/1/0
D       172.16.0.160/27 [90/25625856] via 172.16.1.69, 00:03:19, GigabitEthernet0/3/0
D       172.16.0.192/27 [90/25625856] via 172.16.1.77, 00:03:19, GigabitEthernet0/1
O       172.16.0.224/27 [110/2] via 172.16.1.73, 00:02:30, GigabitEthernet0/2/0
D       172.16.1.0/27 [90/25625856] via 172.16.1.69, 00:03:19, GigabitEthernet0/3/0
O       172.16.1.32/28 [110/2] via 172.16.1.73, 00:02:30, GigabitEthernet0/2/0
C       172.16.1.48/28 is directly connected, GigabitEthernet0/0/0
L       172.16.1.49/32 is directly connected, GigabitEthernet0/0/0
C       172.16.1.64/30 is directly connected, GigabitEthernet0/1/0
L       172.16.1.66/32 is directly connected, GigabitEthernet0/1/0
C       172.16.1.68/30 is directly connected, GigabitEthernet0/3/0
L       172.16.1.70/32 is directly connected, GigabitEthernet0/3/0
C       172.16.1.72/30 is directly connected, GigabitEthernet0/2/0
L       172.16.1.74/32 is directly connected, GigabitEthernet0/2/0
C       172.16.1.76/30 is directly connected, GigabitEthernet0/1
L       172.16.1.78/32 is directly connected, GigabitEthernet0/1
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/28 is directly connected, GigabitEthernet0/0
L       209.165.200.226/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 209.165.200.225
```

-OSPF

```
Main-MLS#SHOW IP ROUTE
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.1.66 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 16 subnets, 5 masks
C       172.16.0.0/26 is directly connected, Vlan100
L       172.16.0.1/32 is directly connected, Vlan100
O E2    172.16.0.64/26 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
C       172.16.0.128/27 is directly connected, Vlan200
L       172.16.0.129/32 is directly connected, Vlan200
O E2    172.16.0.160/27 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O E2    172.16.0.192/27 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O       172.16.0.224/27 [110/3] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O E2    172.16.1.0/27 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O       172.16.1.32/28 [110/3] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O       172.16.1.48/28 [110/2] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
C       172.16.1.64/30 is directly connected, GigabitEthernet1/1/1
L       172.16.1.65/32 is directly connected, GigabitEthernet1/1/1
O E2    172.16.1.68/30 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O       172.16.1.72/30 [110/2] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O E2    172.16.1.76/30 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
     209.165.200.0/28 is subnetted, 1 subnets
O E2    209.165.200.224 [110/20] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
O*E2 0.0.0.0/0 [110/1] via 172.16.1.66, 00:26:43, GigabitEthernet1/1/1
```

```
C:\>PING 172.16.1.37

Pinging 172.16.1.37 with 32 bytes of data:

Reply from 172.16.1.37: bytes=32 time<1ms TTL=125
Reply from 172.16.1.37: bytes=32 time<1ms TTL=125
Reply from 172.16.1.37: bytes=32 time<1ms TTL=125
Reply from 172.16.1.37: bytes=32 time<1ms TTL=125

Ping statistics for 172.16.1.37:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 172.16.1.37

Tracing route to 172.16.1.37 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.0.1
  2   0 ms      0 ms      0 ms      172.16.1.66
  3   0 ms      0 ms      0 ms      172.16.1.73
  4   *         0 ms      0 ms      172.16.1.37

Trace complete.

C:\>
```

(ping from pc1 to pc4 testing OSPF connectivity)

```
S-MLS#SHOW IP ROUTE
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.1.74 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 16 subnets, 5 masks
O       172.16.0.0/26 [110/3] via 172.16.1.74, 00:27:25, GigabitEthernet1/1/1
O E2    172.16.0.64/26 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
O       172.16.0.128/27 [110/3] via 172.16.1.74, 00:27:25, GigabitEthernet1/1/1
O E2    172.16.0.160/27 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
O E2    172.16.0.192/27 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
C       172.16.0.224/27 is directly connected, Vlan400
L       172.16.0.225/32 is directly connected, Vlan400
O E2    172.16.1.0/27 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
C       172.16.1.32/28 is directly connected, Vlan25
L       172.16.1.33/32 is directly connected, Vlan25
O       172.16.1.48/28 [110/2] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
O       172.16.1.64/30 [110/2] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
O E2    172.16.1.68/30 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
C       172.16.1.72/30 is directly connected, GigabitEthernet1/1/1
L       172.16.1.73/32 is directly connected, GigabitEthernet1/1/1
O E2    172.16.1.76/30 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
     209.165.200.0/28 is subnetted, 1 subnets
O E2    209.165.200.224 [110/20] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
O*E2 0.0.0.0/0 [110/1] via 172.16.1.74, 00:27:35, GigabitEthernet1/1/1
```

```
C:\>ping 172.16.0.132

Pinging 172.16.0.132 with 32 bytes of data:

Reply from 172.16.0.132: bytes=32 time<1ms TTL=125
Reply from 172.16.0.132: bytes=32 time<1ms TTL=125
Reply from 172.16.0.132: bytes=32 time<1ms TTL=125
Reply from 172.16.0.132: bytes=32 time<1ms TTL=125

Ping statistics for 172.16.0.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 172.16.0.132

Tracing route to 172.16.0.132 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.0.225
  2   0 ms      0 ms      0 ms      172.16.1.74
  3   0 ms      0 ms      0 ms      172.16.1.65
  4   *         0 ms      1 ms      172.16.0.132

Trace complete.

C:\>
```

(ping from pc6 to pc3 testing OSPF connectivity)

-EIGRP

```
N-MLS#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.1.70 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 16 subnets, 5 masks
D EX    172.16.0.0/26 [170/281856] via 172.16.1.70, 00:31:28, GigabitEthernet1/1/1
D       172.16.0.64/26 [90/25626112] via 172.16.1.70, 00:32:17, GigabitEthernet1/1/1
D EX    172.16.0.128/27 [170/281856] via 172.16.1.70, 00:31:28, GigabitEthernet1/1/1
C       172.16.0.160/27 is directly connected, Vlan600
L       172.16.0.161/32 is directly connected, Vlan600
D       172.16.0.192/27 [90/25626112] via 172.16.1.70, 00:32:17, GigabitEthernet1/1/1
D EX    172.16.0.224/27 [170/281856] via 172.16.1.70, 00:31:28, GigabitEthernet1/1/1
C       172.16.1.0/27 is directly connected, Vlan500
L       172.16.1.1/32 is directly connected, Vlan500
D EX    172.16.1.32/28 [170/281856] via 172.16.1.70, 00:31:28, GigabitEthernet1/1/1
D EX    172.16.1.48/28 [170/281856] via 172.16.1.70, 00:32:17, GigabitEthernet1/1/1
D EX    172.16.1.64/30 [170/281856] via 172.16.1.70, 00:32:18, GigabitEthernet1/1/1
C       172.16.1.68/30 is directly connected, GigabitEthernet1/1/1
L       172.16.1.69/32 is directly connected, GigabitEthernet1/1/1
D       172.16.1.72/30 [90/3072] via 172.16.1.70, 00:32:17, GigabitEthernet1/1/1
D       172.16.1.76/30 [90/3072] via 172.16.1.70, 00:32:17, GigabitEthernet1/1/1
S*   0.0.0.0/0 [1/0] via 172.16.1.70
```

```
C:\>ping 172.16.0.66

Pinging 172.16.0.66 with 32 bytes of data:

Reply from 172.16.0.66: bytes=32 time=9ms TTL=125
Reply from 172.16.0.66: bytes=32 time<1ms TTL=125
Reply from 172.16.0.66: bytes=32 time<1ms TTL=125
Reply from 172.16.0.66: bytes=32 time=1ms TTL=125

Ping statistics for 172.16.0.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

```
C:\>tracert 172.16.0.66

Tracing route to 172.16.0.66 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.1.1
  2   0 ms      0 ms      0 ms      172.16.1.70
  3   0 ms      1 ms      0 ms      172.16.1.77
  4   *         0 ms      0 ms      172.16.0.66

Trace complete.

C:\>
```

(ping from pc7 to pc9 testing EIGRP connectivity)

```
R-MLS#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.1.78 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 16 subnets, 5 masks
D EX    172.16.0.0/26 [170/281856] via 172.16.1.78, 00:34:43, GigabitEthernet1/0/5
C       172.16.0.64/26 is directly connected, Vlan700
L       172.16.0.65/32 is directly connected, Vlan700
D EX    172.16.0.128/27 [170/281856] via 172.16.1.78, 00:34:43, GigabitEthernet1/0/5
D       172.16.0.160/27 [90/25626112] via 172.16.1.78, 00:35:32, GigabitEthernet1/0/5
C       172.16.0.192/27 is directly connected, Vlan800
L       172.16.0.193/32 is directly connected, Vlan800
D EX    172.16.0.224/27 [170/281856] via 172.16.1.78, 00:34:43, GigabitEthernet1/0/5
D       172.16.1.0/27 [90/25626112] via 172.16.1.78, 00:35:32, GigabitEthernet1/0/5
D EX    172.16.1.32/28 [170/281856] via 172.16.1.78, 00:34:43, GigabitEthernet1/0/5
D EX    172.16.1.48/28 [170/281856] via 172.16.1.78, 00:35:32, GigabitEthernet1/0/5
D EX    172.16.1.64/30 [170/281856] via 172.16.1.78, 00:35:32, GigabitEthernet1/0/5
D       172.16.1.68/30 [90/3072] via 172.16.1.78, 00:35:32, GigabitEthernet1/0/5
D       172.16.1.72/30 [90/3072] via 172.16.1.78, 00:35:32, GigabitEthernet1/0/5
C       172.16.1.76/30 is directly connected, GigabitEthernet1/0/5
L       172.16.1.77/32 is directly connected, GigabitEthernet1/0/5
S*   0.0.0.0/0 [1/0] via 172.16.1.78
```

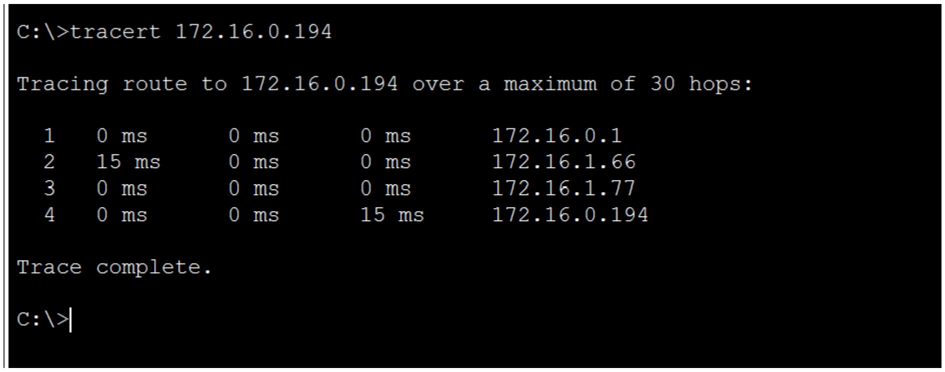
```
C:\>ping 172.16.0.162

Pinging 172.16.0.162 with 32 bytes of data:

Reply from 172.16.0.162: bytes=32 time<1ms TTL=125
Reply from 172.16.0.162: bytes=32 time<1ms TTL=125
Reply from 172.16.0.162: bytes=32 time<1ms TTL=125
Reply from 172.16.0.162: bytes=32 time=1ms TTL=125

Ping statistics for 172.16.0.162:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
C:\>tracert 172.16.0.162

Tracing route to 172.16.0.162 over a maximum of 30 hops:

  1   4294967295 ms0 ms      0 ms      172.16.0.193
  2   0 ms      0 ms      0 ms      172.16.1.78
  3   1 ms      0 ms      0 ms      172.16.1.69
  4   *         0 ms      0 ms      172.16.0.162

Trace complete.

C:\>
```

(ping from pc10 to pc8 testing EIGRP connectivity)

- Redistribution

```
C:\>ping 172.16.0.194

Pinging 172.16.0.194 with 32 bytes of data:

Reply from 172.16.0.194: bytes=32 time<1ms TTL=125
Reply from 172.16.0.194: bytes=32 time<1ms TTL=125
Reply from 172.16.0.194: bytes=32 time=1ms TTL=125
Reply from 172.16.0.194: bytes=32 time=1ms TTL=125

Ping statistics for 172.16.0.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

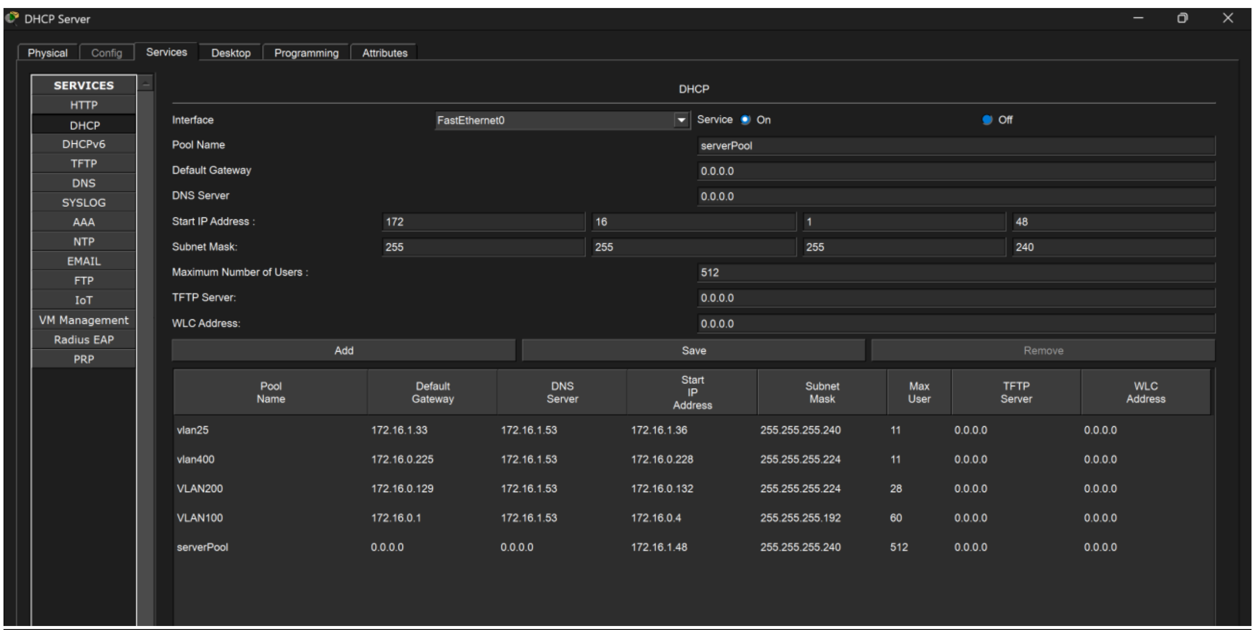


```
C:\>tracert 172.16.0.194

Tracing route to 172.16.0.194 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.0.1
  2   15 ms     0 ms      0 ms      172.16.1.66
  3   0 ms      0 ms      0 ms      172.16.1.77
  4   0 ms      0 ms      15 ms     172.16.0.194

Trace complete.

C:\>
```

(ping from pc1 to pc10 testing redistribution)

-DHCP Server:

Configure DHCP Server on Main Branch and Configure a Branch_GW Router as DHCP Server



-NAT:

Configure Dynamic NAT with PAT and Static NAT.

```
MIU-GW#show running-config | include ip nat
 ip nat outside
 ip nat inside
 ip nat inside
 ip nat inside
 ip nat inside
 ip nat inside
ip nat pool my-pool 209.165.200.227 209.165.200.227 netmask 255.255.255.240
ip nat inside source list 1 pool my-pool overload
ip nat inside source static 172.16.1.52 209.165.200.228
ip nat inside source static 172.16.1.53 209.165.200.229
ip nat inside source static 172.16.1.51 209.165.200.230
```

```
MIU-GW#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
---  209.165.200.228    172.16.1.52        ---                ---
---  209.165.200.229    172.16.1.53        ---                ---
---  209.165.200.230    172.16.1.51        ---                ---
```

```
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 209.165.200.225

Tracing route to 209.165.200.225 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.0.1
  2   0 ms      0 ms      0 ms      172.16.1.66
  3   0 ms      0 ms      0 ms      209.165.200.225

Trace complete.

C:\>
```

(ping from pc1 to ISP testing NAT)

```
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

(ping from pc6 to ISP testing NAT)

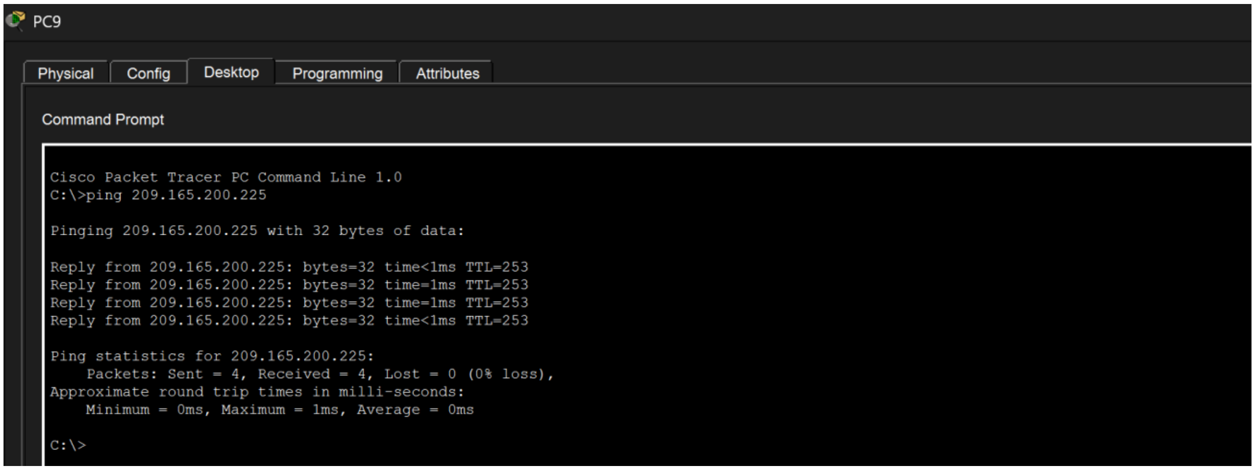
PC8

Physical | Config | Desktop | Programming | Attributes

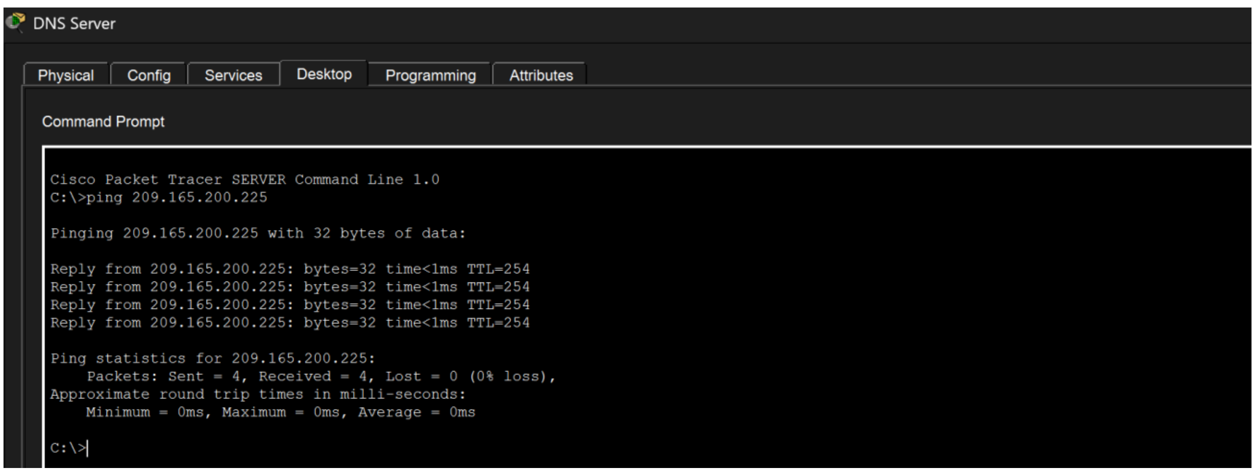
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time<1ms TTL=253
Reply from 209.165.200.225: bytes=32 time=1ms TTL=253

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

(ping from pc8 to ISP testing NAT)



(ping from pc9 to ISP testing NAT)

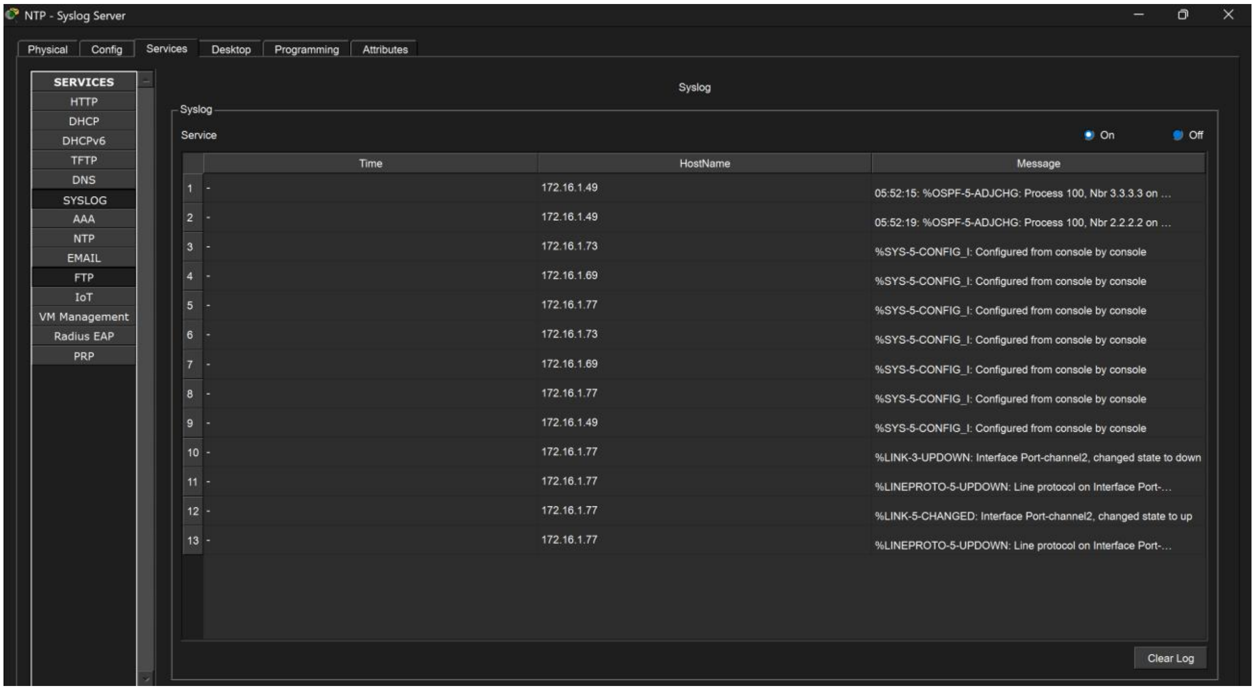


(ping from DNS to ISP testing NAT)

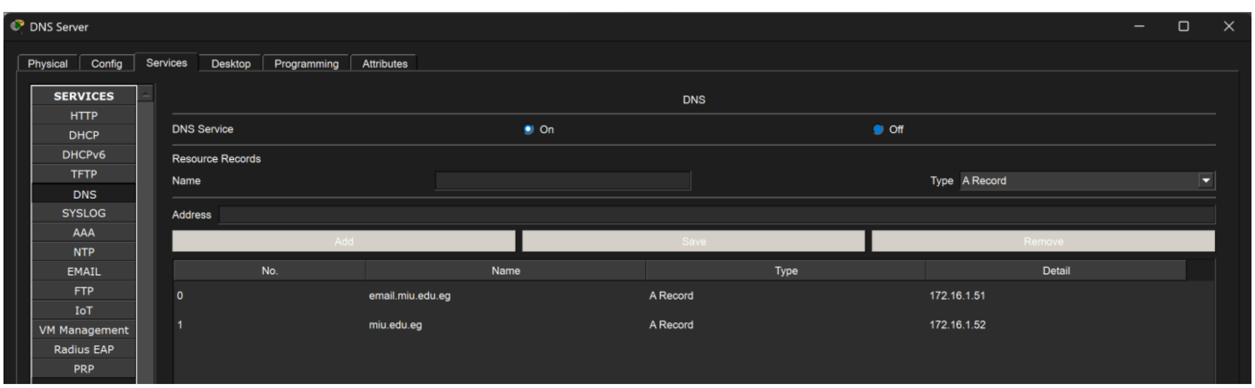
-Network Management Features:

Configure various network management features.

-NTP Server:



-DNS Server




PC1
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL http://miu.edu.eg
Go
Stop
Welcome to Misr International University
MIU - Faculty of Computer Science
CSC230 - Computer Networks
Misr International University
جامعة مصر الدولية


PC9
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL http://miu.edu.eg
Go
Stop
Welcome to Misr International University
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Misr International University
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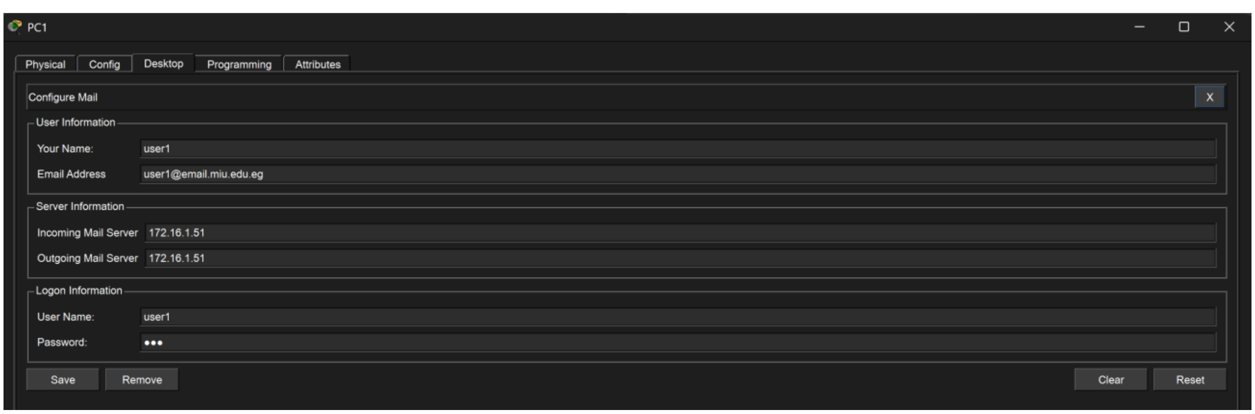
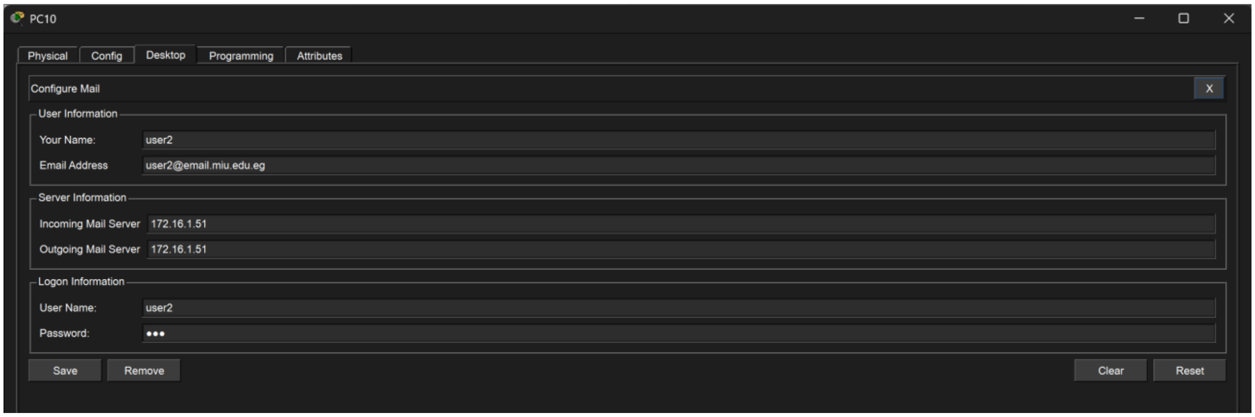
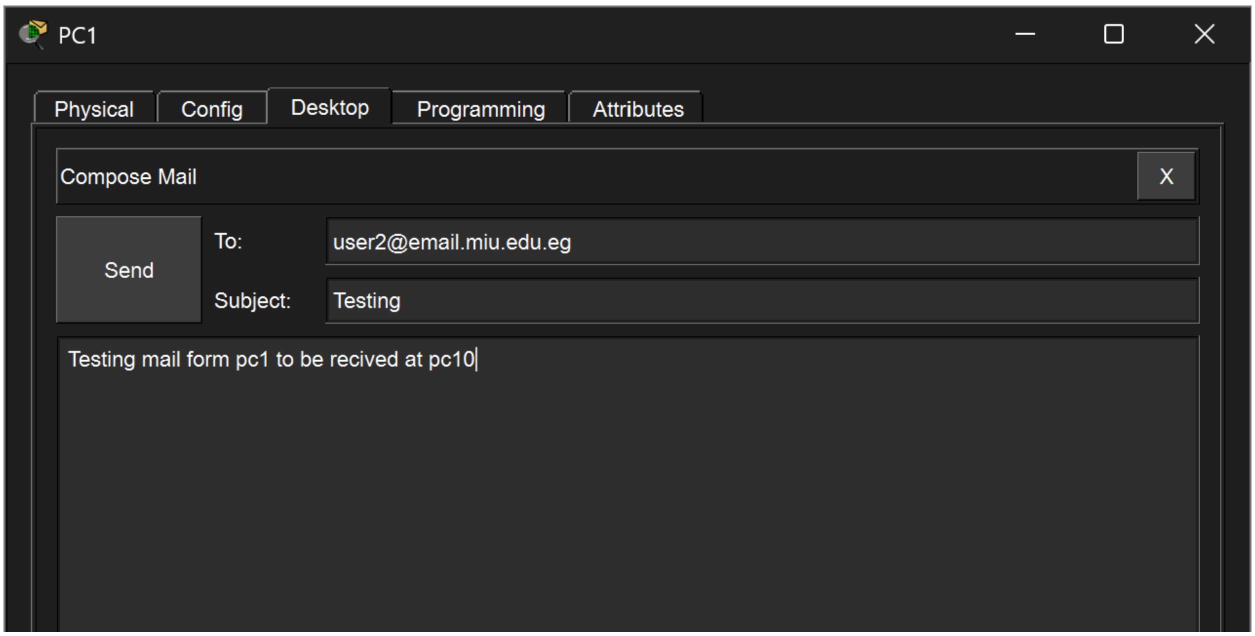
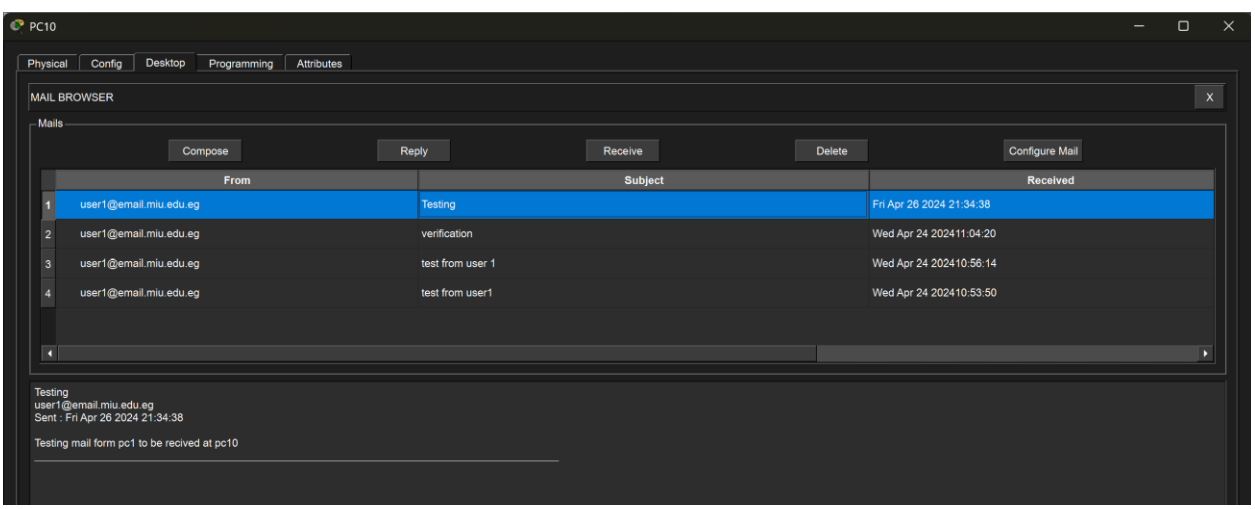
-Email:


PC1
Physical
Config
Desktop
Programming
Attributes
Configure Mail
User Information
Your Name: user1
Email Address user1@email.miu.edu.eg
Server Information
Incoming Mail Server 172.16.1.51
Outgoing Mail Server 172.16.1.51
Logon Information
User Name: user1
Password: •••
Save
Remove
Clear
Reset


PC10
Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name: user2
Email Address user2@email.miu.edu.eg
Server Information
Incoming Mail Server 172.16.1.51
Outgoing Mail Server 172.16.1.51
Logon Information
User Name: user2
Password: •••
Save
Remove
Clear
Reset


PC1
Physical
Config
Desktop
Programming
Attributes
Compose Mail
X
Send
To: user2@email.miu.edu.eg
Subject: Testing
Testing mail form pc1 to be recived at pc10


Sending mail to user2@email.miu.edu.eg , with subject : Testing .. Mail Server: 172.16.1.51
Send Success.
Cancel
Send/Receive


PC10
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
1 user1@email.miu.edu.eg Testing Fri Apr 26 2024 21:34:38
2 user1@email.miu.edu.eg verification Wed Apr 24 2024 11:04:20
3 user1@email.miu.edu.eg test from user 1 Wed Apr 24 2024 10:56:14
4 user1@email.miu.edu.eg test from user1 Wed Apr 24 2024 10:53:50
Testing
user1@email.miu.edu.eg
Sent : Fri Apr 26 2024 21:34:38
Testing mail form pc1 to be recived at pc10

-VPN:

Configure a Site-to-Site IPsec VPN.

```
C:\>ping 192.168.2.6

Pinging 192.168.2.6 with 32 bytes of data:

Reply from 192.168.2.6: bytes=32 time<1ms TTL=125
Reply from 192.168.2.6: bytes=32 time<1ms TTL=125
Reply from 192.168.2.6: bytes=32 time<1ms TTL=125
Reply from 192.168.2.6: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.2.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 192.168.2.6

Tracing route to 192.168.2.6 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.0.1
  2   0 ms      4 ms      0 ms      172.16.1.66
  3   0 ms      0 ms      0 ms      64.100.1.2
  4   0 ms      0 ms      0 ms      192.168.2.6

Trace complete.
```

```
MIU-GW#show crypto ipsec sa

interface: GigabitEthernet0/0
    Crypto map tag: VPN-MAP, local addr 209.165.200.226

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (172.16.0.0/255.255.0.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
   current_peer 64.100.1.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 13, #pkts encrypt: 13, #pkts digest: 0
   #pkts decaps: 9, #pkts decrypt: 9, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 1, #recv errors 0

     local crypto endpt.: 209.165.200.226, remote crypto endpt.:64.100.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
     current outbound spi: 0x1B24A6CC(455386828)

     inbound esp sas:
      spi: 0x0A0B0438(168494136)
        transform: esp-aes 256 esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2008, flow_id: FPGA:1, crypto map: VPN-MAP
        sa timing: remaining key lifetime (k/sec): (4525504/3438)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0x1B24A6CC(455386828)
        transform: esp-aes 256 esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2009, flow_id: FPGA:1, crypto map: VPN-MAP
```

```
      sa timing: remaining key lifetime (k/sec): (4525504/3438)
      IV size: 16 bytes
      replay detection support: N
      Status: ACTIVE

   outbound ah sas:

   outbound pcp sas:
```

-Wireless Home Router:

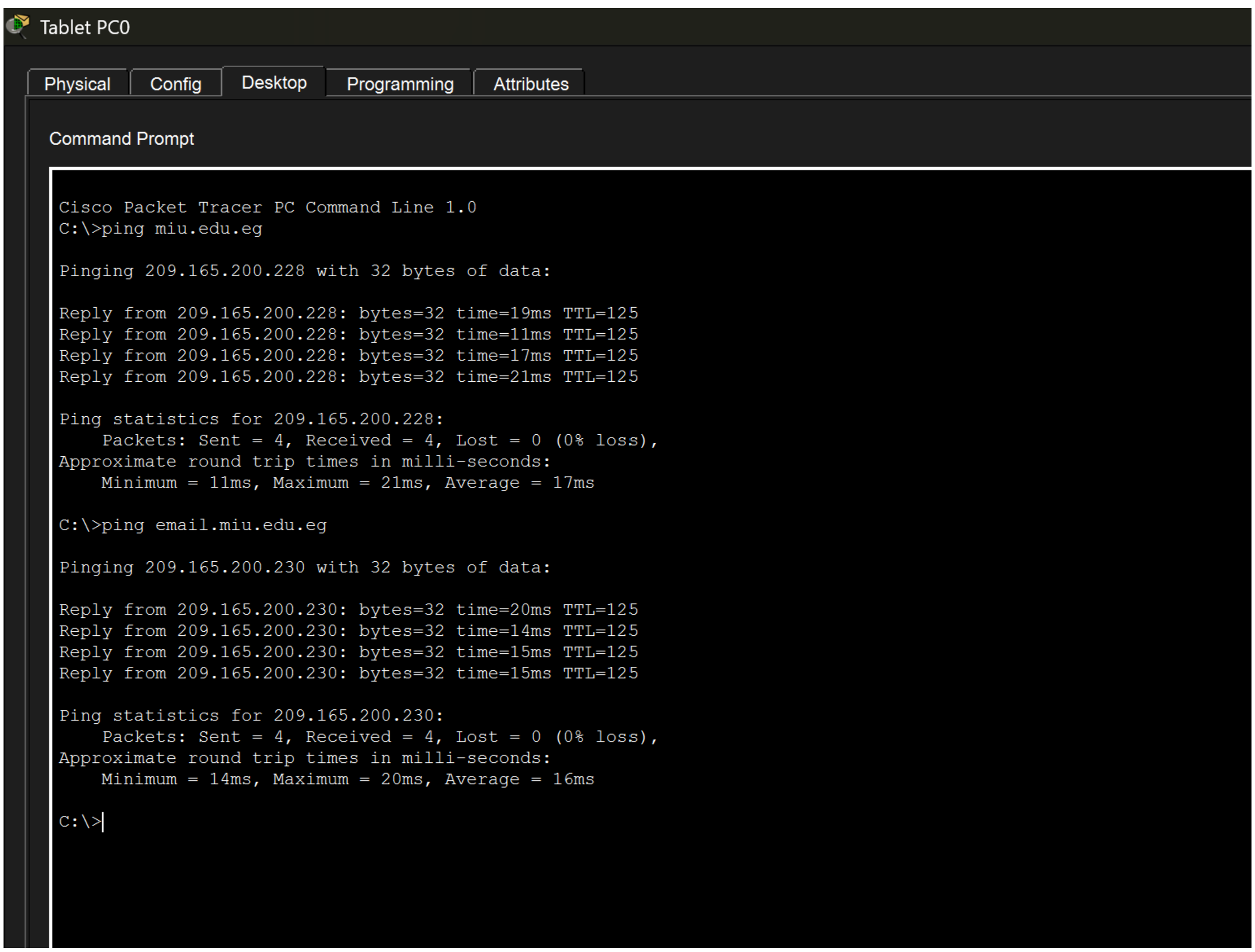
Configure a Wireless Home Router.

```
C:\>ping 209.165.200.229

Pinging 209.165.200.229 with 32 bytes of data:

Reply from 209.165.200.229: bytes=32 time<1ms TTL=125
Reply from 209.165.200.229: bytes=32 time<1ms TTL=125
Reply from 209.165.200.229: bytes=32 time<1ms TTL=125
Reply from 209.165.200.229: bytes=32 time<1ms TTL=125

Ping statistics for 209.165.200.229:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

(ping from Laptop0 to DNS)

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.C900.D5A6
   Link-local IPv6 Address.........: FE80::201:C9FF:FE00:D5A6
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.2
   Subnet Mask.....................: 255.255.255.128
   Default Gateway.................: ::
                                     192.168.10.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-56-B9-9C-08-00-01-C9-00-D5-A6
   DNS Servers.....................: ::
                                     209.165.200.229

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.433D.A95C
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-56-B9-9C-08-00-01-C9-00-D5-A6
   DNS Servers.....................: ::
                                     209.165.200.229
```


Laptop0
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://miu.edu.eg
Go
Stop
Welcome to Misr International University
MIU - Faculty of Computer Science
CSC230 - Computer Networks
Misr International University
جامعة مصر الدولية


Tablet PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping miu.edu.eg
Pinging 209.165.200.228 with 32 bytes of data:
Reply from 209.165.200.228: bytes=32 time=19ms TTL=125
Reply from 209.165.200.228: bytes=32 time=11ms TTL=125
Reply from 209.165.200.228: bytes=32 time=17ms TTL=125
Reply from 209.165.200.228: bytes=32 time=21ms TTL=125
Ping statistics for 209.165.200.228:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 11ms, Maximum = 21ms, Average = 17ms
C:\>ping email.miu.edu.eg
Pinging 209.165.200.230 with 32 bytes of data:
Reply from 209.165.200.230: bytes=32 time=20ms TTL=125
Reply from 209.165.200.230: bytes=32 time=14ms TTL=125
Reply from 209.165.200.230: bytes=32 time=15ms TTL=125
Reply from 209.165.200.230: bytes=32 time=15ms TTL=125
Ping statistics for 209.165.200.230:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 14ms, Maximum = 20ms, Average = 16ms
C:\>